

Atishaya Maharjan

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EDUCATION

BSc. Joint Honours: Computer Science & Mathematics, Minor in Statistics Aug. 2022 – Present
University of Manitoba *Winnipeg, Canada*

International Undergraduate Student Scholarship | Dean's Honour Roll

PROFESSIONAL EXPERIENCE

Software Engineer Mar. 2024 – Present
LG Electronics *Englewood Cliffs, USA (Remote)*

- Shipped **Next.js/Redux/TypeScript** B2B platform features for **1,000+** vendors, contributing to a **25%** user-engagement lift.
- Designed analytics and tagging frameworks adopted by **6+** product teams, enabling real-time measurement and data-driven feature prioritization.
- Built AI-powered global search with **35 ms** lower latency and checkout workflows that drove a **15%** conversion lift across promo-code and magic-link flows.

Software Engineer
Vibecore – Cloud Infrastructure & CI/CD

- Automated CI/CD deployments from **GitHub Actions** to **AWS EC2**, standardizing application releases.
- Configured **S3**, **DynamoDB**, and **IAM** permissions for cloud-connected application workflows.
- Provisioned and maintained **EBS/EFS** storage for application data and deployment infrastructure.

Software Engineer Jun. 2025 – Jan. 2026
Hoek Insights *Amsterdam, Netherlands (Remote)*

- Engineered a **Python** package for **Three.js** visualizations in **Jupyter**, reducing render time by **38%** and bundle size by **25%** through **React** renderer optimization.
- Shipped a **FastAPI/PostgreSQL** MVP with **18+** endpoints and **10** product workflows.
- Maintained below **200 ms** staging latency and **95%** integration-test coverage with automated API tests.

Software Engineer Aug. 2019 – Apr. 2024
Timeero *Phoenix, USA (Remote)*

- Led frontend development of a **TypeScript/React/Vite/Mapbox** mileage-tracking application supporting **1,000+** active users.
- Shipped **15+** geospatial workflows for segment mapping, mileage capture, and field operations.
- Improved time-capture accuracy by **30%** through Mapbox-based mapping and production workflow refinements.

SELECTED PROJECTS

PayTime | *Next.js, FastAPI, Supabase* **1st Place**

- Won **1st place** at the CSSA Design & Craft Case Competition 2026 by building an auditable payroll-ingestion platform.
- Implemented field mapping, validation, exception review, and source traceability for **CSV**, **XLSX**, **PDF**, image, and email-derived records.
- Built and pitched the full-stack prototype in a **6.5-hour** sprint using **Next.js**, **TypeScript**, **FastAPI**, **Python**, and **Supabase**.

HAPA | *Next.js, React, TypeScript, AI/LLM* **Winner**

- Won the **AccelR8 Prairie AI Hackathon 2026** with a working AI shopping prototype that learns preferences and personalizes product feeds.
- Combined onboarding signals, swipe-based feedback, and natural-language/voice pivots to continuously adapt recommendations.
- Built and pitched the prototype in a **12-hour** sprint with an extensible architecture for multimodal personalization and AI-assisted commerce.

- WinLocate
Full-Stack, Open Data, AI
1st Place

 - Built a full-stack data platform using **Socrata** open data for companies considering relocation to Winnipeg.
 - Implemented city comparisons, zoning maps, budget tools, and relocation-planning workflows for applied civic data analysis.
 - Won **1st place** among **15** teams at the Winnipeg Open Data Datathon 2026.
- Brainstorm
Full-Stack, RAG, Data Systems

 - Built a role-aware knowledge-management platform integrating **Jira**, **PostgreSQL**, **Redis**, and **Pinecone** through retrieval-augmented generation.
 - Designed role-based retrieval and source grounding across **4** connected systems, restricting responses to authorized organizational data.
 - Implemented multi-model routing with dynamic selection and fallback policies balancing latency, reliability, and response quality.

SELECTED RESEARCH EXPERIENCE

- Algorithms & Computational Geometry
GADA Lab
Mar. 2024 – Present

 - Optimizing collaborative drone-delivery routes for logistics by developing exact and approximation algorithms with dynamic programming, exchange-point geometry, and structural analysis.
 - Developed an approximation algorithm for Hamiltonian-path optimization in spatially embedded graphs, with visualization and randomized testing infrastructure for **100+** synthetic instances.
 - Improved the performance of a noncrossing spanning-tree research library by **30%** and introduced stronger **Python** development practices.
- Machine Learning Research
Terrabyte Machine Learning Research Group
May 2025 – Present

 - Extended diffusion-based learning to higher-order relational data by generalizing Graph Neural Diffusion architectures to hypergraphs.
 - Implemented **4** Hypergraph Neural Diffusion variants for graph and hypergraph machine-learning experiments.
 - Built a reproducible **MLflow** pipeline tracking **25+** experiments across **5+** datasets.
- Combinatorics & Graph Theory
University of Manitoba
Jan. 2024 – Jul. 2025

 - Developed a symbolic and computational proof framework for the Hurlbert–Kamat property on perfect m -ary trees.
 - Derived a closed-form expression for the independence number of perfect m -ary trees using generating functions and recurrence analysis.
 - Used induction and **SageMath** validation to test conjectures and verify symbolic identities across **1,000+** instances.

TECHNICAL SKILLS

- Languages:

TypeScript, JavaScript, Python, C++, Java, SQL, Bash
- Frameworks & Libraries:

React, Next.js, Redux, FastAPI, Node.js, PyTorch, Three.js
- Data:

PostgreSQL, Redis, MongoDB, DynamoDB, Pinecone
- Cloud & Infrastructure:

AWS EC2, S3, IAM, EBS/EFS, Docker, GitHub Actions, Linux
- Engineering Practices:

CI/CD, API design, integration testing, automated testing, reproducible experimentation
- Specializations:

Full-stack development, cloud infrastructure, ML systems, geospatial systems

SELECTED AWARDS

- CSSA Design & Craft Case Competition – 1st Place

2026
- AccelR8 Prairie AI Hackathon – Winner

2026
- CSSA Capture the Flag – 1st of 41 teams (2026), 2nd of 25+ teams (2025)
- Winnipeg Open Data Datathon – 1st Place among 15 teams

2026
- S.U.R.E. Poster Competition – 3rd Place, Computational Science

Aug. 2024

COMMUNITY & MENTORSHIP

- CS Help Center Leader & Teaching Assistant:

mentored **100+** students and graded **150+** algorithms assignments;
- Coding Kickoff Facilitator:

taught Git and IDE setup to **50+** students;
- .devClub Events Executive:

co-led technical workshops and hackathons.